Andrew C. Photinakis

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EDUCATION

Rochester Institute of Technology — Rochester, NY
Master of Science in Computer Science — Expected December 2026
Bachelor of Science in Computer Science — December 2025
Minor in Finance – Awards/Honors: RIT Presidential Scholar & Dean's List

SKILLS & LANGUAGES

- **Languages:** Rust, C++, C, Java, Python, C#, TypeScript, SQL, Bash
- **Distributed Systems & Networking:** Tokio, Tonic, gRPC, Protocol Buffers, MPI, OpenMP, Raft, UDP, TCP/IP, SMTP, POP3
- **AI/ML & Scientific Computing:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Qiskit Aer, Cirq
- **Backend & Web:** Spring Boot, .NET Core, FastAPI, Flask, Node.js, React, Angular, Next.js
- **Cloud & Infrastructure:** AWS, Azure Functions, Azure Data Factory, GCP, GitLab CI, CMake, Docker
- **Databases:** PostgreSQL, MongoDB, BigQuery, MySQL, DynamoDB

WORK EXPERIENCE

Amazon Web Services (AWS) – *Software Development Engineer Intern* — May 2026 – August 2026

- Engineered highly scalable public-facing APIs and asynchronous backend orchestration workflows for cloud networking infrastructure, enabling automated, low-latency resource provisioning across multiple distributed environments.
- Developed a control-plane service to surface real-time network routing health, utilizing optimized database views and custom state-reconciliation algorithms to enhance customer network visibility and diagnostic capabilities.
- Accelerated team engineering workflows by **30%** by architecting a multi-agent Generative AI orchestration system, leveraging Model Context Protocol (MCP) servers and Large Language Models to automate task research, code generation, and test planning.

PricewaterhouseCoopers (PwC) – *Software Engineer & Cyber Risk Consulting Intern* — June 2025 – August 2025

- Accelerated AI-driven document processing pipeline by **2x** through consolidating Python and .NET workflows into a unified C# (.NET Core) system, improving reliability and maintainability for LLM orchestration and YAML validation.
- Boosted dashboard refresh speed and accuracy by designing a modular Angular frontend integrated with Node.js and BigQuery, enabling real-time metric visualizations in Google Cloud Console for faster business insights.
- Reduced manual setup time by **90%** with a cross-platform Avalonia UI automation tool and cut Vertex AI YAML hallucinations by **80%** by implementing robust prompt-safe schema constraints, enhancing overall system stability and user trust.

TechSource, Inc. – *Software Engineering Co-op* — January 2024 – December 2025

- Led development of a fully automated Java data pipeline on Azure Function Apps, slashing execution time by **98%**, improving data accuracy and flexibility, reducing operational costs, and automating daily execution reports sent to business stakeholders, enhancing transparency and operational oversight.
- Integrated Swagger and SOAP APIs with modular Java libraries to achieve **99%** data transfer accuracy and streamline complex data extraction and transformation workflows, enabling more reliable downstream analytics.
- Created reusable, responsive UI components with **TypeScript**, **TailwindCSS**, and **Next.js** for the U.S. Department of Energy's GPT-4-powered AI platform, enhancing user experience and boosting frontend performance.

PROJECTS

Standardized Quantum Benchmarking Framework – *Rust, Qiskit Aer, Cirq, GitLab CI* — August 2026 – Present

- Engineered a vendor-neutral benchmarking platform executing 3 quantum algorithms across 12 test cases and 2 simulation frameworks using a Rust control plane with process-isolated Qiskit Aer and Cirq adapters over a versioned JSON protocol.
- Established reproducible scientific validation with protocol fixtures and 200+ automated tests by comparing results against independent analytic and exact-evolution references while recording accuracy, latency, resource usage, seeds, and cryptographic provenance.
- Implemented an end-to-end quantum benchmarking stack spanning all quantum stack layers with fail-closed contracts, cross-layer validation, multidimensional performance metrics, and SPEC-inspired anti-gaming controls.

Raft-Replicated Order Matching and Execution Sandbox – *Rust, Tokio, Tonic, TLA+* — August 2026 – Present

- Achieved identical order, fill, and portfolio state across a 3-node Raft cluster through leader failover by implementing deterministic state-machine replication, replicated request deduplication, and canonical state-digest verification.
- Validated crash recovery and effectively-once order semantics across 10 injected fault scenarios, including leader crashes, partitions, message loss/reordering/duplication, and follower restarts, using a seeded discrete-event simulator and checksummed WAL.

SBMPI Blockchain Simulator – *C++17, MPI, OpenMP, PBFT, secp256k1, CMake* — October 2025 – December 2025

- Engineered a high-performance distributed blockchain simulator in C++17 using MPI to parallelize transaction validation and consensus across independent shards, leveraging a hierarchical architecture of Shard Leaders and a Final Committee to achieve horizontal scalability.
- Implemented the **Practical Byzantine Fault Tolerance (PBFT)** consensus algorithm with a 3-phase commit protocol and quorum enforcement, ensuring system liveness and safety while coordinating block finalization across multiple process-isolated committees.
- Parallelized transaction verification within individual nodes using OpenMP and the secp256k1 library, significantly reducing block validation latency and increasing throughput under high-frequency transaction loads.